

and responsibility: during the great depression of the 1920s in the US, when the average company would tend to either shut shop or exploit cheap labour, IBM actually provided jobs and life insurance—and even paid vacations!

And then, later, there was WWII: at that time (from the IBM archives), “all IBM facilities were placed at the disposal of the U.S. government. IBM’s product line expanded to include bombsights, rifles and engine parts—in all, more than three dozen major ordinance items. Thomas Watson, Sr., set a nominal one percent profit on those products and used the money to establish a fund for widows and orphans of IBM war casualties.”

For IBM’s history, visit www-03.ibm.com/ibm/history/.

1931 / 1935:

The company introduced the IBM 601 in one of these two years, depending on whom you ask; it was a punch card machine with an arithmetic unit based on “relays.” It could do a multiplication in one second. It read two numbers up to eight decimal places from a card and punched the result on a blank space on the same card.

(A relay is an electrical switch that opens and closes under the control of another electrical circuit. Note that it is still “electrical,” not “electronic.”)

The machine became important in scientific as well as commercial computation, with several hundred units manufactured. Remember that computers or computing machines at the time weren’t counted in the millions!

1.8 The Universe Opens Up

1937

Alan Mathison Turing of Cambridge published a paper on “computable numbers,” which was, essentially, the mathematical theo-